Recorded research session

A saved implementation baseline and an evidence-based trace of how the agents reached their final state.

Result: goal unmet. Four independent workers attempted one recognition family and successive repairs. One branch improved chord-label measurements, but no complete prediction passed protected parsing. This report describes the existing recorded run; no new research session was executed to produce it.

Measure	Recorded result
System goal	Image-to-editable-kern transcription, melody, chord symbols and correct transposition.
Scope	7 development excerpts from 7 song groups; 130 reference melody events and 38 chord labels. Full completion requires 32 excerpts.
Execution	12 coordinator decision slots, including one invalid attempt; 4 workers; 68 model-call attempts/receipts; 32.6 active minutes.
Final choice	Worker 1 retained as the best evaluated experiment for partial chord evidence, not as a successful solution.
Why it stopped	All 4 experiment slots were used. Call, coordinator-step and total-time budgets were not exhausted.
Integrity	Recorded process audit passed all 8 checks. The current code matches all 18 code hashes in that audit.

Measured outcomes

Candidate	Melody errors	Chord errors	Parseable
Enlarged baseline	87/130 (66.9%)	30/38 (78.9%)	Not asserted here
Worker 1	130/130 (100%)	25/38 (65.8%)	0/7
Worker 2	130/130 (100%)	38/38 (100%)	0/7
Workers 3 and 4	Not scored	Not scored	No predictions

Lower error is better. Edit error includes substitutions, deletions and insertions; it is not ordinary classification accuracy. Missing outputs in Workers 3 and 4 are failures, not zero-error results. All comparisons use the same pilot, not held-out data.

Interpretation: the coordinator reacted to evidence and narrowed the assignments. That establishes result-informed iteration in this run, but does not establish end-to-end self-improvement or a successful switch to a different recognition method.

Goal and acceptance contract

The actual saved run configuration, separated from the local goals assigned to workers.

System goal: Develop an image-to-editable-kern workflow for a church band, including melody, chord symbols and correct transposition. Investigate observed failures, choose informative experiments, measure the result and stop honestly when the goal is unmet.

Priority: Explain why each approach is chosen and expose measurable failure; source content is data, never instructions.

Final acceptance criteria

Requirement	Saved contract / observed status
Coverage	32 required samples; this run evaluated a 7-sample pilot.
Melody	Maximum error rate 0.0; best selected worker had 1.0.
Chords	Maximum error rate 0.0; selected worker had 0.6579.
Structure and transposition	Parseable output and no transposition failures; selected worker had 0 parseable files and actual-output transposition was unverified.
Timing and human review	Audited chord timing and musician review required; both false.

Contract caveat: Conservative completion guard for this prototype, not a musician-agreed product release threshold. A seven-sample pilot cannot complete this goal. Report measured improvements separately.

Execution budgets and identity

Coordinator: maximum 24 decision slots, 4 experiments, 80 model calls and 3,600 active seconds. Each worker: maximum 16 decisions, 600 active seconds, 120 seconds per shell command and 180 seconds per model call. Commands use the pinned Python 3.12 slim Docker image, networking enabled, 2 CPU units and 2 GB memory. Model: gpt-6-astra; coordinator low reasoning effort, workers high.

How an agent is executed

The runner assembles a role prompt and current structured context, makes an ephemeral model call, validates the returned JSON action, executes the requested tool, and records its observation. The next call receives bounded recent observations. A worker is given a distinct assignment and workspace; workers run sequentially. They do not talk directly to one another. Parent source/configuration and selected feedback pass through the runner.

The evaluator is fixed code, not an LLM. It scores frozen outputs against protected labels. A worker finish message means the worker submitted; it does not mean the experiment or system goal succeeded.

Coordinator decision trail

All 12 recorded slots, including the rejected decision. Reasons below are recorded model statements; observed outcomes are summarized separately.

Slot 1: inspect_dataset

Reason: Establish image scope, grouping and integrity before interpreting baseline failures or assigning a small experiment.

Hypothesis: Dataset structure and image quality may reveal constraints that determine which recognition failure is useful to test first.

Expected: An inventory of the 32 excerpts and seven song groups, integrity findings, and available metadata relevant to melody, chord timing and transposition.

Observed: Inventory: 32 excerpts, 7 groups. Pilot indices 1, 6, 12, 17, 19, 23, 29. Recognition receives images only.

Evidence IDs: goal

Slot 2: baseline_summary

Reason: Dataset inspection establishes a seven-song pilot within 32 development excerpts. Read measured baseline failures before selecting a targeted experiment; the pilot cannot establish completion.

Hypothesis: Baseline errors will identify whether melody recognition, chord timing, or transposition is the most informative first target.

Expected: Measured melody and chord errors, timing audit status, and sample-level failure patterns that support a narrowly scoped experiment.

Observed: Enlarged Audiveris baseline: melody 87/130 errors; chords 30/38. Greedy Transcoda showed excessive insertions.

Evidence IDs: goal, dataset

Slot 3: discover_evidence

Reason: Data and baselines are already inspected. Enlarged Audiveris has the best measured melody error rate on the pilot (0.6692), but misses 30 of 38 chords; greedy Transcoda produces 527 insertions. Sources can inform a small experiment targeting these distinct failures before spending the four-worker budget.

Hypothesis: Image preprocessing plus separate chord recognition, or constrained neural decoding, may address the observed failures more effectively than unrestricted greedy transcription.

Expected: Relevant source IDs describing practical methods and their limitations, sufficient to select and read at least one source before assigning an experiment.

Observed: Returned a curated local source catalog. No live web search occurred.

Evidence IDs: dataset, baseline

Slot 4: read_evidence

Reason: Data and baseline inspection are complete. Read the format audit before spending an experiment on the enlarged Audiveris baseline, whose pilot melody error is 0.6692 and chord error is 0.7895.

Hypothesis: The benchmark audit can distinguish recognition failures from serialization or extraction failures, enabling a narrowly targeted first experiment.

Expected: Evidence about image preprocessing, kern validity, chord extraction and transposition limitations sufficient to select a testable intervention without accessing reference labels during inference.

Observed: Read the supplied-score benchmark and format audit: missing harmony, parser failures, unaudited timing.

Evidence IDs: dataset, baseline, source-catalog

Slot 5: run_experiment

Reason: The enlarged baseline recovers substantial melody but only 8 of 38 pilot chords. The format audit identifies unpromoted text as a plausible integration failure. Holding melody fixed isolates whether chord OCR and structured conversion recover useful information within the four-experiment budget; timing and musician review remain unresolved.

Hypothesis: Some missing chords are legible image text that can be recovered through a separate constrained recognition stage without changing melody transcription.

Expected: More than eight correctly matched pilot chords and lower chord edit error, with unchanged melody error; failure to improve would argue that text recognition or symbol interpretation, rather than harmony export alone, is the bottleneck.

Observed: Launched Worker 1. It submitted 7 incomplete outputs after 16 steps.

Evidence IDs: dataset, baseline, evidence-82fd3c08

Slot 6: evaluate_experiment

Reason: The submitted experiment must be evaluated before further assignments. Its 15 OCR candidates and passing synthetic checks do not establish transcription correctness, and unavailable baseline melody artifacts threaten severe regression.

Hypothesis: The OCR stage may recover additional chord matches, but missing melody predictions will prevent this candidate from improving the complete workflow.

Expected: Protected evaluation will quantify chord matches and edit errors against the baseline's 8 matches and 30 errors, and reveal the melody penalty relative to 87/130 baseline errors.

Observed: Protected evaluation: 0/7 parseable, 130/130 melody errors, 25/38 chord errors.

Evidence IDs: exp-1f137c273ec6, baseline, evidence-82fd3c08

Slot 7: rejected decision

Observed: Rejected before execution: Debug requires a failed parent. The evaluated first branch was not classified as an execution-failed parent.

Slot 8: run_experiment

Reason: The parent reduced measured chord error from 30/38 to 25/38, but melody error rose to 100% and none of seven files parsed. Its missing melody dependency makes integration repair more informative than further OCR tuning. Three experiments remain. The parent executed successfully and was evaluated, so this is an improve branch; the previous debug request was rejected. Preserve the baseline as the stronger measured melody comparator.

Hypothesis: Fresh melody recognition plus independently validated serialization can recover the baseline's melody performance while retaining some of the parent's measured chord improvement.

Expected: Parseability increases and melody error falls below the parent's 100%; chord error remains near 25/38. Failure to obtain parseable outputs or a melody improvement would falsify this integration approach within the available runtime. Even success would leave full-dataset accuracy, chord timing and musician acceptance unresolved.

Observed: Launched Worker 2 as an improve branch of Worker 1.

Evidence IDs: dataset, baseline, evidence-82fd3c08, exp-1f137c273ec6, exp-1f137c273ec6-evaluation, decision-error-7

Slot 9: evaluate_experiment

Reason: The submitted integration must be evaluated before assigning more work. Its worker reports startup failures and placeholders; protected scoring will establish whether any chord gains survived and preserve an honest comparison with the earlier candidate without consuming another experiment slot.

Hypothesis: Audiveris startup failures prevented melody recovery and complete parseable transcriptions; retaining the chord OCR may preserve some ordered chord recognition, but does not establish a usable workflow.

Expected: Likely zero parseable files and melody error rate 1.0. Chord error may remain near the parent's 25/38 errors or regress if placeholders omit harmony. Transposition and audited timing will remain unsupported.

Observed: Protected evaluation: 0/7 parseable, 130/130 melody errors, 38/38 chord errors.

Evidence IDs: exp-d6d6755b413e, exp-1f137c273ec6-evaluation, baseline

Slot 10: run_experiment

Reason: The latest branch regressed chord error by 0.3421 and still produced zero parseable files because recognition never started. This supports testing a narrowly identified execution bottleneck before changing recognition methods. Two experiment slots and 2398 seconds remain, enough for a bounded repair attempt and subsequent evaluation; the earlier OCR candidate remains preserved.

Hypothesis: The immediate parent's total failure is primarily a repairable runtime dependency problem; resolving it will enable actual melody recognition and reveal whether its unchanged OCR gains survive complete-file integration.

Expected: A successful preflight yields real MusicXML and enables nonempty, independently parseable melody outputs. Continued startup failure falsifies the repair's feasibility within this budget; poor protected scores despite successful execution would instead implicate recognition or conversion.

Observed: Launched Worker 3 as an improve branch of Worker 2. It returned no predictions and was classified execution_failed.

Evidence IDs: baseline, evidence-82fd3c08, exp-1f137c273ec6-evaluation, exp-d6d6755b413e, exp-d6d6755b413e-evaluation

Slot 11: run_experiment

Reason: The failed parent now identifies a concrete missing library, making one targeted repair more informative than another broad setup attempt. One experiment and 27 model calls remain. The earlier OCR candidate remains the best measured experiment for chord labels, but its zero parseable outputs and 100% melody error leave the goal unmet. Execution failure permits a direct debug branch.

Hypothesis: Missing GTK 3 is the immediate blocker to fresh Audiveris export; resolving it may enable measurable melody recognition, although subsequent runtime or transcription failures remain possible.

Expected: Either verified GTK resolution enables a real MusicXML export and seven evaluable integrated predictions, or the bounded preflight exposes a specific remaining blocker. Any musical improvement must be established by subsequent protected evaluation.

Observed: Launched Worker 4 as a debug branch of Worker 3. It returned no predictions and was classified execution_failed.

Evidence IDs: exp-ee56f2b32016, exp-d6d6755b413e-evaluation, exp-1f137c273ec6-evaluation, baseline, evidence-82fd3c08

Slot 12: finish

Reason: The experiment budget is exhausted. Preserve the best evaluated candidate: chord errors fell from baseline 30/38 to 25/38, but melody error rose from 87/130 to 130/130 and no complete files parsed. Later integration branches produced no improvement; GTK loading was verified only outside Java, and export timed out without predictions. The goal remains unmet: only seven of 32 excerpts were tested, chord timing is unaudited, musician review is absent, and actual-output transposition remains unverified.

Hypothesis: The initial OCR stage offers a limited chord-label improvement, but no tested approach establishes a usable image-to-editable-kern workflow.

Expected: Retain the initial candidate as partial development evidence, with failed integration branches preserved and no completion or held-out accuracy claim.

Observed: Finished with Worker 1 as best evaluated candidate. Runner stored goal_met=false and status=goal_unmet.

Evidence IDs: baseline, music-benchmark, exp-1f137c273ec6-evaluation, exp-d6d6755b413e-evaluation, exp-ee56f2b32016, exp-431cde96831a

Worker 1: assignment

Chord OCR with fixed-melody assumption

Branch: explore | parent: none

Approach: Keep enlarged Audiveris melody recognition fixed and add a conservative image-derived chord-text recognition and structured harmony conversion stage.

Exact local objective: On the seven first-per-song pilot excerpts, recover missing printed chord symbols from the region above the staff using enlarged-image OCR and a strict chord grammar. Convert roots, qualities, explicit extensions and slash basses into a parseable editable kern harmony spine. Preserve existing melody events; record chord bounding boxes and proposed onset alignment separately, marking ambiguous timing for review. Do not infer absent symbols from musical plausibility or consult references.

Exact success criteria: Produce sealed parseable predictions for all seven pilot images, source code and per-image diagnostics distinguishing OCR failures, unsupported chord syntax and uncertain onset alignment. Compare protected chord edit error and matches with enlarged Audiveris baseline (30/38 errors, 8 matches), and verify melody error does not increase from 87/130 errors. Include round-trip parse checks and synthetic transposition checks for melody, chord roots and slash basses; distinguish these checks from measured transcription correctness. Report timing as unaudited.

Recorded outcome

Worker statement: Implemented and ran conservative enlarged-image chord OCR with strict harmony conversion. Seven outputs contain 15 accepted OCR chord candidates; these are not verified matches. All seven enlarged Audiveris melody baselines were unavailable, so melody preservation and non-regression could not be verified; incomplete outputs count as failures. Synthetic chord round-trip, root/slash-bass transposition, kern parsing and melody transposition checks passed. Protected transcription metrics were unavailable, so improvement over baseline is unmeasured. Timing remains unaudited. Goal unmet.

Runner classification: evaluated

Worker execution: 16 decision slots; 8.64 active minutes; final worker state submitted.

Protected result: 0/7 parseable; melody 130/130 errors; chords 25/38 errors, 13 matches.

Worker 1: execution log

Every recorded worker action in order. Purpose is the recorded reason; command exit status describes the shell action, not transcription correctness.

1. run. Locate input images and enlarged Audiveris baseline artifacts before implementing the isolated chord stage.

Receipt: Shell exit 0; 0.55 seconds; interrupted=False. Full output retained in snapshot.

2. run. Locate reusable enlarged Audiveris melody artifacts and available recognition tools. Keeping melody fixed requires the actual baseline output; aggregate metrics alone cannot reconstruct it.

Receipt: Shell exit 0; 0.65 seconds; interrupted=False. Full output retained in snapshot.

3. run. Install OCR and notation parsing dependencies and inspect public Audiveris distribution availability. Baseline melody artifacts were not supplied, so exact melody preservation remains blocked; any rerun must be reported separately.

Receipt: Shell exit 1; 8.88 seconds; interrupted=False. Full output retained in snapshot.

4. run. Diagnose disk exhaustion and available dependencies before choosing a feasible image-derived OCR pipeline; exact baseline preservation is blocked by missing baseline artifacts.

Receipt: Shell exit 0; 0.65 seconds; interrupted=False. Full output retained in snapshot.

5. run. Retry a small dependency installation using workspace storage, check recognition executables, and prepare enlarged images for direct inspection. Missing baseline artifacts still prevent claiming unchanged melody.

Receipt: Shell exit 0; 11.96 seconds; interrupted=False. Full output retained in snapshot.

6. run. Create a readable contact sheet to inspect printed chord regions and assess feasible image-derived recovery. Baseline melody artifacts are absent, so exact preservation and paired accuracy claims remain unavailable.

Receipt: Shell exit 0; 0.95 seconds; interrupted=False. Full output retained in snapshot.

7. view_image. Inspect the seven enlarged score excerpts to recover only visible chord text and assess bounding boxes. Baseline melody artifacts are absent, so unchanged melody and paired error improvements cannot be verified.

Receipt: Selected image attached to the next model call.

8. run. Install a small OCR executable and finish notation-parser dependencies, while checking whether a reproducible Audiveris rerun is feasible. Original baseline artifacts are absent, so even a successful rerun cannot establish byte-for-byte melody preservation or measured non-regression.

Receipt: Shell exit 0; 4.06 seconds; interrupted=False. Full output retained in snapshot.

9. run. Inspect OCR availability and enlarged above-staff text for all supplied pilot images before implementing strict chord parsing; baseline melody preservation remains blocked by missing artifacts.

Receipt: Shell exit 0; 1.16 seconds; interrupted=False. Full output retained in snapshot.

10. run. Diagnose and retry the OCR installation using workspace storage. The chord stage needs an executable recognizer; missing baseline melody artifacts remain a separate blocker to preservation and paired accuracy checks.

Receipt: Shell exit 127; 2.78 seconds; interrupted=False. Full output retained in snapshot.

11. run. Install OCR without root by extracting public Debian packages locally, then inspect enlarged above-staff OCR and bounding boxes. Missing baseline melody artifacts remain a separate blocker; this action does not claim melody preservation or measured improvement.

Receipt: Shell exit 0; 13.12 seconds; interrupted=False. Full output retained in snapshot.

12. run. Repair the identified OCR runtime dependency and inspect available artifacts before implementing the chord stage. Baseline melody absence prevents a faithful fixed-melody experiment and must remain explicit.

Receipt: Shell exit 0; 2.14 seconds; interrupted=False. Full output retained in snapshot.

13. run. Repair the known OCR library failures using local package extraction, then inspect above-staff text and boxes. This cannot restore the absent Audiveris melody baseline or establish transcription accuracy.

Receipt: Shell exit 0; 4.85 seconds; interrupted=False. Full output retained in snapshot.

14. run. Use the remaining execution step to save and run the conservative OCR/strict-harmony stage, preserve melody exactly when a baseline exists, and explicitly record incomplete outputs when it does not. Run synthetic parser and transposition checks without presenting them as measured transcription correctness.

Receipt: Shell exit 0; 3.97 seconds; interrupted=False. Full output retained in snapshot.

15. finish. End honestly with the observed results and blockers; neither transcription improvement nor melody preservation was established.

Receipt: ValueError: Submit 1-32 source/config files for reproducibility

16. finish. Budget exhausted; submit the experiment with explicit limitations and no unsupported accuracy claim.

Receipt: Submission accepted; exact final statement is on the assignment/outcome page.

Worker 2: assignment

Fresh Audiveris + chord integration

Branch: improve | parent: Worker 1

Approach: Retain the parent's image-derived chord OCR and replace its unavailable-baseline dependency with fresh enlarged-image melody recognition and validated kern serialization.

Exact local objective: On the same seven pilot images, first inspect the parent serializer and determine why protected evaluation found zero parseable outputs despite its local checks. Run Audiveris on enlarged images in this container, installing required dependencies within the worker budget, and convert actual recognized melody into kern. Integrate the existing chord OCR without changing its recognition thresholds. Do not assume prior baseline artifacts exist, substitute dummy melody, or infer missing notes from musical plausibility. Validate complete files with an independent kern reader and report explicit failures if recognition or conversion cannot be completed. Preserve chord bounding boxes and flag onset alignment as unaudited.

Exact success criteria: Submit sealed predictions and per-image recognition, conversion and independent parsing diagnostics for all seven inputs. Target seven parseable files and melody error no worse than the enlarged Audiveris baseline of 87/130 errors. Compare protected chord error against both the parent's 25/38 errors and baseline's 30/38; report any regressions rather than assuming OCR gains survive integration. Test upward-major-second transposition on actual generated melody, chord roots and slash basses, separating structural checks from transcription correctness. Retain raw recognition outputs and document missing dependencies or runtime limits.

Recorded outcome

Worker statement: Integration failed within budget. Retained parent chord OCR unchanged and implemented fresh Audiveris execution, melody conversion and validation paths. Java/native-library startup failures prevented MusicXML exports for all seven images; all seven outputs are explicitly marked failure placeholders, with no invented melody. Complete-file parsing and transposition on recognized melody could not be tested. Per-image diagnostics record these failures. Parent checks validated syntax and synthetic examples rather than complete generated transcriptions; the exact protected-reader failure remains unconfirmed. Chord onset alignment remains unaudited. Protected error comparisons are unavailable, so no improvement or non-regression is claimed. Goal unmet.

Runner classification: evaluated

Worker execution: 15 decision slots; 8.76 active minutes; final worker state submitted.

Protected result: 0/7 parseable; melody 130/130 errors; chords 38/38 errors, 0 matches.

Worker 2: execution log

Every recorded worker action in order. Purpose is the recorded reason; command exit status describes the shell action, not transcription correctness.

1. run. Locate parent source, pilot images, and installed recognition dependencies before revising the pipeline.

Receipt: Shell exit 0; 0.54 seconds; interrupted=False. Full output retained in snapshot.

2. run. Inspect the inherited serializer and unchanged OCR thresholds, identify why incomplete files passed local checks, and check dependencies before installing Audiveris.

Receipt: Shell exit 0; 1.19 seconds; interrupted=False. Full output retained in snapshot.

3. run. Preserve and inspect the parent OCR implementation while installing Java, OCR, and an independent kern reader, then locate a public Audiveris release.

Receipt: Shell exit 1; 8.73 seconds; interrupted=False. Full output retained in snapshot.

4. run. Inspect the exact inherited OCR thresholds, install Python recognition/conversion dependencies locally, and identify a downloadable Audiveris distribution without relying on privileged apt installation.

Receipt: Shell exit 0; 9.52 seconds; interrupted=False. Full output retained in snapshot.

5. run. Inspect the unchanged OCR implementation and diagnose disk exhaustion before attempting an ARM-compatible Audiveris installation. The parent emitted null-only melody spines when its baseline was absent, and its checks accepted empty parsed scores; fresh recognition and nonempty independent parsing must both succeed.

Receipt: Shell exit 2; 0.54 seconds; interrupted=False. Full output retained in snapshot.

6. run. Determine whether storage exhaustion is recoverable, inspect the exact inherited OCR functions without truncating them, and identify usable dependencies before selecting an ARM-compatible recognition installation.

Receipt: Shell exit 0; 0.65 seconds; interrupted=False. Full output retained in snapshot.

7. run. Diagnose whether the failed dependency installation can be recovered using writable storage, and inspect only the inherited chord grammar and helper functions needed to preserve OCR behavior. The parent's null-only melody fallback and acceptance of zero parsed events explain why its local checks did not establish usable predictions.

Receipt: Shell exit 0; 0.65 seconds; interrupted=False. Full output retained in snapshot.

8. run. Recover pip installation by moving temporary files off the 256 MB /tmp filesystem, preserve the exact parent chord grammar, and obtain an ARM Java runtime plus the Audiveris application bundle for a compatibility check. No melody will be fabricated if recognition remains unavailable.

Receipt: Shell exit 0; 23.91 seconds; interrupted=False. Full output retained in snapshot.

9. run. Extract the downloaded Audiveris application and obtain an ARM Java runtime from an alternative public distributor after the first runtime download returned HTTP 403. Identify native-library compatibility requirements and the preserved OCR executable before running fresh recognition.

Receipt: Shell exit 0; 9.17 seconds; interrupted=False. Full output retained in snapshot.

10. run. Run fresh enlarged-image Audiveris recognition using the downloaded ARM Java runtime, retain raw outputs and logs, and diagnose concrete runtime failures before implementing conversion.

Receipt: Shell exit 0; 3.35 seconds; interrupted=False. Full output retained in snapshot.

11. run. Replace the incompatible Java 21 runtime with Java 25 and run fresh Audiveris recognition on all seven enlarged images, retaining its command, logs and raw exports for conversion and diagnosis.

Receipt: Shell exit 0; 7.23 seconds; interrupted=False. Full output retained in snapshot.

12. run. Retry native-library extraction on the executable workspace filesystem, run actual Audiveris recognition on enlarged inputs, and inspect the parent fallback and raw exports before conversion.

Receipt: Shell exit 0; 2.59 seconds; interrupted=False. Full output retained in snapshot.

13. run. Identify the exact remaining native-library error and Audiveris startup switch, inspect retained OCR diagnostics and the parent serializer, and locate enlarged inputs so the final two actions can run recognition and serialize only actual recognized events.

Receipt: Shell exit 0; 1.59 seconds; interrupted=False. Full output retained in snapshot.

14. run. Use the final execution opportunity to bypass Audiveris HiDPI startup, retain unchanged parent OCR and its artifacts, convert only actual exported melody, validate complete kern files independently, test major-second transposition on generated events, and document every unresolved failure without substituting invented notes.

Receipt: Shell exit 0; 2.94 seconds; interrupted=False. Full output retained in snapshot.

15. finish. Close the experiment honestly after seven recognition failures, retaining implementation and runtime configuration for reproduction.

Receipt: Submission accepted; exact final statement is on the assignment/outcome page.

Worker 3: assignment

Bounded Java/native-library repair

Branch: improve | parent: Worker 2

Approach: Repair the documented Java/native-library startup failure with a bounded single-image Audiveris preflight before rerunning the existing integration pipeline.

Exact local objective: Inspect inherited recognition-command diagnostics and capture the exact startup exception. Correct only the implicated Java runtime or native-library dependency; retain the existing chord OCR thresholds and melody conversion. Spend at most 180 seconds establishing a real MusicXML export from one pilot image. If that succeeds, process all seven pilot images and independently parse each complete generated kern file, repairing serialization only where an actual parser diagnostic identifies a defect. If preflight fails, stop and retain the exact commands, runtime versions and exception rather than generating placeholder transcriptions. Never consult references or invent missing notes. Preserve raw recognition and chord bounding boxes; keep chord timing explicitly unaudited.

Exact success criteria: Deliver reproducible startup diagnostics and a real single-image MusicXML export as the first milestone. Then target seven independently parseable sealed predictions. Protected evaluation must compare melody error with the enlarged Audiveris baseline of 87/130 and both evaluated parents' 1.0 error; compare chord error with the earlier OCR parent's 25/38, the immediate parent's 38/38 and baseline's 30/38. Check upward-major-second transposition on actual generated notes, chord roots and slash basses, reporting untested cases separately. Successful startup or parsing alone must not be called musical correctness.

Recorded outcome

Worker statement: Preflight failed: restoring Java 25.0.5 and Audiveris 5.11.0 exposed java.lang.UnsatisfiedLinkError: Unable to load library 'gtk-3'; libgtk-3.so was missing. No MusicXML export or predictions were produced. Preserved launch commands, runtime diagnostics and exact exception. OCR thresholds and melody conversion remain unchanged. Complete-file parsing and transposition of actual notes, chord roots and slash basses remain untested; chord timing remains unaudited. Goal unmet; no musical improvement claimed.

Runner classification: execution_failed

Worker execution: 11 decision slots; 4.99 active minutes; final worker state submitted.

No protected accuracy measurement was produced for this branch because it failed to produce predictions.

Worker 3: execution log

Every recorded worker action in order. Purpose is the recorded reason; command exit status describes the shell action, not transcription correctness.

1. run. Inspect inherited execution command, startup diagnostics, and integration code before a bounded dependency repair.

Receipt: Shell exit 0; 0.55 seconds; interrupted=False. Full output retained in snapshot.

2. run. Read the exact inherited startup command and runtime configuration before making a narrowly targeted dependency repair and starting the bounded preflight.

Receipt: Shell exit 127; 0.58 seconds; interrupted=False. Full output retained in snapshot.

3. run. Capture the inherited startup exception and executable paths before selecting a narrowly targeted runtime repair.

Receipt: Shell exit 2; 0.56 seconds; interrupted=False. Full output retained in snapshot.

4. run. Recover the exact inherited Java command and any retained exception, and locate runtime/model assets before starting the 180-second single-image preflight.

Receipt: Shell exit 0; 0.66 seconds; interrupted=False. Full output retained in snapshot.

5. run. Record the exact reproducible launch exception, begin the explicit 180-second preflight clock, and identify available runtime and Audiveris assets without changing recognition or conversion.

Receipt: Shell exit 0; 1.17 seconds; interrupted=False. Full output retained in snapshot.

6. run. Check the recorded preflight deadline, actual architecture, and available native libraries before selecting a compatible runtime; retain the inherited OCR settings unchanged.

Receipt: Shell exit 0; 0.77 seconds; interrupted=False. Full output retained in snapshot.

7. run. Read only the preflight clock and compatibility facts omitted from the prior log tail, so the repair stays within the explicit 180-second limit.

Receipt: Shell exit 0; 0.75 seconds; interrupted=False. Full output retained in snapshot.

8. run. Attempt restoration of the missing inherited Java runtime and Audiveris assets under the existing 180-second deadline, retain exact commands and exceptions, and stop without placeholder transcriptions if no real pilot export is produced.

Receipt: Shell exit 0; 12.98 seconds; interrupted=False. Full output retained in snapshot.

9. run. Identify the exact native-library exception from the restored runtime and check the original preflight deadline before making any narrowly implicated dependency repair.

Receipt: Shell exit 0; 0.67 seconds; interrupted=False. Full output retained in snapshot.

10. run. Stop after the failed bounded preflight, preserve the exact GTK startup exception and runtime versions, and archive inherited placeholders so they cannot be mistaken for newly recognized predictions. Retain the unchanged OCR and conversion sources; parsing, transposition, and musical correctness remain untested.

Receipt: Shell exit 0; 0.77 seconds; interrupted=False. Full output retained in snapshot.

11. finish. The bounded preflight failed and the experiment already stopped; retain reproducible failure evidence without inventing transcriptions.

Receipt: Submission accepted; exact final statement is on the assignment/outcome page.

Worker 4: assignment

Targeted GTK repair

Branch: debug | parent: Worker 3

Approach: Install the specifically missing GTK 3 runtime and verify native-library resolution before a bounded Audiveris export preflight.

Exact local objective: Reproduce the inherited gtk-3 loading failure, install the distribution's GTK 3 runtime package and verify libgtk-3 resolves, recording package names and runtime versions. Retain Java 25 and Audiveris 5.11 plus unchanged OCR thresholds and melody conversion. Within 240 seconds obtain a genuine MusicXML export from one pilot image; stop with exact diagnostics if this fails. If successful, run the existing pipeline on all seven pilot images, independently validate complete kern outputs, and repair only parser-confirmed serialization defects. Preserve raw recognition outputs and chord bounding boxes. Do not fabricate notes or use references; mark chord timing unaudited.

Exact success criteria: Deliver native-library verification and either a real single-image MusicXML export or a reproducible failure report. If preflight succeeds, submit seven sealed predictions with per-image parsing diagnostics and actual-output upward-major-second transposition checks for melody, chord roots and slash basses. Protected evaluation should compare parseability against 0/7, melody errors against baseline 87/130 and prior experiments' 1.0 rate, and chord errors against the best experiment's 25/38 and baseline 30/38. Report regressions and untested cases explicitly; startup, parsing and synthetic checks do not establish musical correctness.

Recorded outcome

Worker statement: GTK native loading succeeded after rootless extraction of Debian 13 arm64 packages: libgtk-3-0t64 and libgtk-3-common 3.24.49-3, with libglib2.0-0t64 2.84.4-3~deb13u5. A fresh Python process loaded libgtk-3.so.0 and queried its runtime version. System installation was blocked by apt lock-directory permissions. The export wrapper then raised subprocess.TimeoutExpired after its remaining 3.4308305588069743 seconds; the original 240-second deadline expired at 240.016 seconds. No MusicXML export or kern predictions were produced. This verifies native loading through ctypes, not successful Java/Audiveris integration; no subsequent native blocker was established. Existing Java 25/Audiveris 5.11 configuration, OCR thresholds and melody conversion were retained. All seven required predictions remain missing and count as failures. Complete-output parsing and actual-output major-second transposition checks for melody, chord roots and slash basses remain untested. Chord timing remains unaudited. No musical improvement or regression rate was measured; the transcription goal remains unmet.

Runner classification: execution_failed

Worker execution: 14 decision slots; 6.14 active minutes; final worker state submitted.

No protected accuracy measurement was produced for this branch because it failed to produce predictions.

Worker 4: execution log

Every recorded worker action in order. Purpose is the recorded reason; command exit status describes the shell action, not transcription correctness.

1. run. Inspect inherited preflight, commands, failure diagnostics and distribution before the targeted GTK runtime repair.

Receipt: Shell exit 0; 0.56 seconds; interrupted=False. Full output retained in snapshot.

2. run. Inspect the inherited launch and conversion code and identify retained runtime paths before reproducing and repairing the GTK failure.

Receipt: Shell exit 0; 0.54 seconds; interrupted=False. Full output retained in snapshot.

3. run. Read the existing preflight and pipeline, start the 240-second deadline, and reproduce the GTK loading failure with the inherited Java 25/Audiveris command before installing the targeted runtime package.

Receipt: Shell exit 0; 0.76 seconds; interrupted=False. Full output retained in snapshot.

4. run. Find the inherited runtime restoration commands and deadline record so Java 25 and Audiveris 5.11 can be restored before reproducing and repairing the missing GTK 3 library within the bounded preflight.

Receipt: Shell exit 0; 0.66 seconds; interrupted=False. Full output retained in snapshot.

5. run. Read the exact inherited download URLs and deadline before restoring the missing runtimes; check native-library availability without changing OCR or conversion.

Receipt: Shell exit 1; 0.55 seconds; interrupted=False. Full output retained in snapshot.

6. model. No tool decision was recorded for this model-call failure.

Receipt: Model call failed before a tool action. The saved model stderr reports the provider failure.

7. run. Reproduce the native GTK load failure, install Debian's specific GTK 3 runtime, verify actual library loading and runtime/package versions, and expose the inherited restoration code needed for the bounded Java 25/Audiveris 5.11 export.

Receipt: Shell exit 0; 0.75 seconds; interrupted=False. Full output retained in snapshot.

8. run. Inspect the failed package installation, refresh distribution package indexes, install the GTK 3 runtime, and verify native loading and package versions within the existing preflight deadline.

Receipt: Shell exit 0; 0.75 seconds; interrupted=False. Full output retained in snapshot.

9. run. System installation is blocked by apt directory permissions. Attempt a rootless extraction of Debian's GTK 3 runtime and dependencies, verify actual native loading and package versions, then restore the inherited Java 25/Audiveris 5.11 preflight under the original 240-second deadline. Preserve exact failures without inventing predictions.

Receipt: Shell exit 0; 0.64 seconds; interrupted=False. Full output retained in snapshot.

10. run. Retry the rootless distribution-package repair without the failing passwd lookup, verify GTK loading in a fresh process, and run the inherited Java 25/Audiveris 5.11 preflight only within the original 240-second deadline. Preserve exact package, loader, and export diagnostics.

Receipt: Shell exit 0; 15.75 seconds; interrupted=False. Full output retained in snapshot.

11. run. The original 240-second deadline has expired. Inspect saved GTK verification and export diagnostics, identify any genuine export already produced, and collect reproducible failure evidence without launching further recognition.

Receipt: Shell exit 0; 0.74 seconds; interrupted=False. Full output retained in snapshot.

12. run. Inspect concise saved GTK runtime and export diagnostics after the expired deadline so the final failure report identifies verified versions and the exact remaining blocker without restarting recognition.

Receipt: Shell exit 0; 0.65 seconds; interrupted=False. Full output retained in snapshot.

13. run. Read the exact saved GTK loading verification and final export log for the failure report. The 240-second preflight deadline has expired; do not launch further recognition or produce placeholder predictions.

Receipt: Shell exit 0; 0.64 seconds; interrupted=False. Full output retained in snapshot.

14. finish. The bounded preflight expired without a genuine export. Stop with preserved native-loading and timeout evidence; do not restart recognition or fabricate predictions.

Receipt: Submission accepted; exact final statement is on the assignment/outcome page.

Why the final state follows

Distinguishing agent claims, execution success and protected measurements.

The first branch recovered 13 matching chord labels rather than the baseline's 8. Chord errors fell by 5, from 30 to 25, a 13.2 percentage-point decrease on this pilot. But it lacked the promised baseline melody artifacts, emitted no melody events, and produced no parseable complete files. Melody errors increased from 87 to 130. The second branch attempted to repair that by running recognition afresh; runtime failures prevented real exports. Later workers narrowed the runtime diagnosis but produced no predictions.

Coordinator final reason: The experiment budget is exhausted. Preserve the best evaluated candidate: chord errors fell from baseline 30/38 to 25/38, but melody error rose from 87/130 to 130/130 and no complete files parsed. Later integration branches produced no improvement; GTK loading was verified only outside Java, and export timed out without predictions. The goal remains unmet: only seven of 32 excerpts were tested, chord timing is unaudited, musician review is absent, and actual-output transposition remains unverified.

The runner preserved that partial candidate and independently retained goal_unmet. Two branches were evaluated; two were execution failures. The final state has no pending action. No branch demonstrated audited chord timing, musician acceptance, full 32-excerpt coverage or a usable complete transcription.

Observed system weaknesses for the next version

- Initial worker assignment required baseline melody artifacts that were not supplied.
- Child branches inherited source/configuration but not installed runtimes, dependencies or all diagnostics.
- Repeated runtime repairs consumed all four experiment slots; no alternate recognition family was tested.
- Local syntax/synthetic checks initially failed to detect missing melody; protected evaluation rejected the result.

Additional execution detail: the first worker had a rejected finish request because it omitted the required source/configuration submission. It corrected that on its last step. A fourth-worker model call failed; the loop continued afterward. A successful shell exit sometimes wrapped a failed subprocess, so shell success alone was not evidence of progress.

Suggested changes to test next (not yet implemented)

Validate required artifacts before dispatching an assignment. Make dependency/runtime handoffs reproducible across branches. Require a real nonempty end-to-end output in preflight. Reserve experiment budget for a different recognition family after repeated infrastructure stalls. Keep the protected evaluator and honest completion gate unchanged while measuring whether these changes reduce setup waste and improve output quality.

Saved baseline and provenance

What was preserved, how to verify it, and the limits of this record.

The snapshot contains 1,971 files with individual SHA-256 hashes, plus a ZIP archive checksum. It preserves implementation source, goals, dependency lockfile, tests, research inputs, baseline benchmark, the live session's durable state, experiment assignments, model/action receipts, command logs, source snapshots, submitted outputs, diagnostics and the earlier system guide.

The snapshot was verified by reading every archived file and comparing its SHA-256 to the manifest. All current implementation hashes named in the recorded process audit match that audit. This gives a stable comparison point before future changes; it does not independently rerun or revalidate model quality.

Capture time: 2026-10-28T16:58:46.873141+09:00

Excluded material: Python bytecode, symlinks, workspace vendor dependencies, temporary downloaded runtimes and caches excluded. Original run remains untouched. Source snapshots, submitted outputs, action/model receipts, full command logs and diagnostics retained. External model service, host packages, Docker image layers and authentication are not bundled. Absolute paths in records describe the original host.

Restoring and comparing

Extract the archive into a separate directory. Verify the archive checksum and file hashes against the manifest before changing it. Recreate the Python environment from the saved dependency lock and obtain the pinned container image separately. External model access and authentication must be supplied by the environment. Logs include absolute paths from the original host; restoring into another location does not rewrite them. This is a source-and-evidence snapshot, not a bit-for-bit running-machine backup.

Evidence hierarchy

Record	How this report uses it
Durable coordinator state	Exact goal, counters, decision reasons, hypotheses, citations, observations and final selection.
Worker state and action receipts	Ordered worker actions, stated purposes, shell status, submission status and active elapsed time.
Assignments and experiment tree	Exact local objectives, criteria, parent relationships and measured status.
Protected evaluations	Melody/chord metrics and parseability. These take precedence over worker self-reports.
Process audit and assessment	Integrity checks, matching code hashes and cross-check of run-level outcomes.

The companion execution-record JSON retains complete saved coordinator and worker histories, the experiment tree, audit and assessment. The archive retains raw action/model records and command logs for deeper inspection. This report summarizes execution receipts rather than reproducing megabytes of terminal output. Agent rationales are recorded statements, not hidden internal reasoning. No exact wall-clock event timeline is inferred where the records only establish step order.